

SANDESH PRASAD PAUDEL

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[Google Scholar](#) | [GitHub](#) | [LinkedIn](#)

RESEARCH PROFILE

Master of Data Science student with a Computer Engineering background, working on interpretable and causal machine learning for longitudinal clinical data. Lead contributor to a supervised capstone on early prediction of chronic kidney disease (CKD) in diabetic patients (sliding-window ML/DL, SHAP and causal inference); independent study of radiomics-deep learning fusion for mammographic lesion classification with patient-grouped evaluation; co-author of an arXiv preprint on deep-learning OCR for Nepali documents. Applying for a PhD in medical AI / health informatics.

EDUCATION

Master of Information Technology, Charles Darwin University 2025 – 2026

- **Capstone (supervisor Dr Sami Azam):** *Beyond Prediction: Longitudinal Machine Learning and Causal Inference for CKD Risk in Diabetic Patients*, final report submitted June 2026.

Bachelor of Engineering in Computer Engineering, Tribhuvan University, Institute of Engineering 2019 – 2024

- Final-year project: *Mero Nagarikta* (Nepali citizenship-card data extraction), supervised by Er. Nabin Lamichhane; resulted in an arXiv preprint.

PUBLICATIONS

- Dhakal, S., Sigdel, S., **Paudel, S. P.**, Ranabhat, S. K., Lamichhane, N. (2024). Mero Nagarikta: Advanced Nepali Citizenship Data Extractor with Deep Learning-Powered Text Detection and OCR. *arXiv:2410.05721*. (All authors contributed equally.)

RESEARCH PROJECTS

Longitudinal ML and causal inference for CKD risk in diabetic patients Capstone, 2026

Public 10-year cohort of 400 diabetic patients (Mendeley Data). Code: github.com/sandeshpau81/CKD-risk-prediction-ML-DL. Lead contributor (group of four): designed and implemented the data-processing pipeline and all predictive models.

- Built a leakage-aware pipeline: five-year sliding windows predicting CKD in the following year, patient-level 70/15/15 split, scaling and SMOTE fitted on training data only, decision thresholds chosen on validation data only.
- Compared Random Forest, XGBoost, LSTM and BiLSTM with attention, using positive class weighting versus SMOTE. Class-weighted XGBoost performed best: ROC-AUC 0.874 with full history, and ROC-AUC 0.730 with recall 0.80 in the stricter pre-onset-only setting (small test set).
- Engineered trajectory features (slope, variability, acceleration, recency weighting); adding them gave modest gains in the pre-onset setting (XGBoost ROC-AUC 0.730 to 0.743).
- Explained predictions with SHAP and attention weights: BMI change, diabetes duration and insulin use ranked highest, and attention moved from the most recent year to the earliest year once CKD history was removed.
- Estimated causal effects of four exposures (DAG-guided backdoor adjustment; propensity matching, regression and IPW; refutation tests). A high-BMI trajectory had the largest robust effect (ATE about +0.54); poor diet regulation had a sizeable effect (about +0.31) despite low SHAP importance. Flagged a positivity violation for diabetes duration and residual confounding as limits.
- Clustered 184 CKD converters (k-means) into fast BMI-driven and slow comorbidity-driven phenotypes (silhouette 0.19, so treated as exploratory).

Mammographic radiomics and deep learning 2026

Benign vs malignant classification of 2,437 CBIS-DDSM lesion ROIs (Kaggle images matched to public PyRadiomics features). Code: github.com/sandeshpau81/cbis-ddsm-radiomics-deep-learning.

- Designed a leakage-safe protocol: patient-grouped 80/20 train/test split, model selection by 5-fold patient-grouped CV on training patients only, preprocessing and feature selection (LASSO, RFE) inside CV pipelines, fine-tuning on training patients only.
- Compared radiomics, five frozen ImageNet backbones (ResNet50, DenseNet121, EfficientNetV2-S, Swin-T, DeiT-S), partial fine-tuning, and early/late radiomics-deep fusion with XGBoost; reported patient-level bootstrap 95% CIs and paired bootstrap tests.
- On the held-out test set, deep features beat radiomics (DenseNet121 AUC 0.775 vs 0.736); the best model, DenseNet121 + radiomics, reached AUC 0.782 (95% CI 0.735 to 0.828). Early-fusion gains over deep features alone were small (+0.007 to +0.028) and not statistically significant.

Mero Nagarikta: Nepali citizenship-card data extraction

B.E. final-year project, 2024

Group of five, equal contribution. **My role:** backend, covering model training and the inference API used by the mobile app.

- Fine-tuned YOLOv8 (transfer learning) to detect text fields on 250 annotated card images (210 train / 40 validation): mAP 99.1% on the front and 96.1% on the back at confidence 0.5.
- Built the Django REST inference API: receives base64-encoded card images, pre-processes them (grayscale, denoising, edge detection, perspective correction), detects fields with YOLOv8, reads them with Nepali-optimised PyTesseract, applies fuzzy-match post-correction (RapidFuzz) and returns structured JSON to the Flutter app.

APPLIED MACHINE LEARNING

Air-quality pollution-event detection (AWS), Machine Learning Engineer

Ongoing

Responsible for the layer-2 pollution-event detection stage of a medallion (bronze/silver/gold) data pipeline on AWS (S3, Glue, Athena, Lambda, IAM).

- Performed EDA on silver-layer sensor data and engineered temporal, station-baseline and multi-sensor features.
- Fitted an unsupervised anomaly-detection ensemble (Isolation Forest, Local Outlier Factor, DBSCAN), evaluated the models, and wrote alert scores to the gold table for downstream reporting.

Other projects

- **ASK-Appliance:** retrieval-augmented assistant (LlamaIndex, vector database) that answers appliance questions grounded in manuals.
- **NSW road-crash analytics:** three-page Power BI dashboard on NSW Government crash data.

TECHNICAL SKILLS

- **Machine learning:** Python, PyTorch, Tensorflow, scikit-learn, XGBoost; deep learning for sequences and images; anomaly detection; transformers and RAG systems
- **Research methods:** interpretable ML (SHAP), causal inference, leakage-safe evaluation, bootstrap statistics
- **Medical imaging and vision:** radiomics, CNN and transformer models, object detection and OCR
- **Data engineering and cloud:** SQL, AWS (S3, Glue, Athena, Lambda, IAM), Git/GitHub

EXPERIENCE

Freelance software developer

2020 – present

- Designed, built and delivered 10+ commercial websites and apps for clients.

Referees: available on request.